

Puzzles to Puzzle You (Remastered)

Welcome to the seventh iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Moving Sofa Problem (Variation)) and pose the next one.

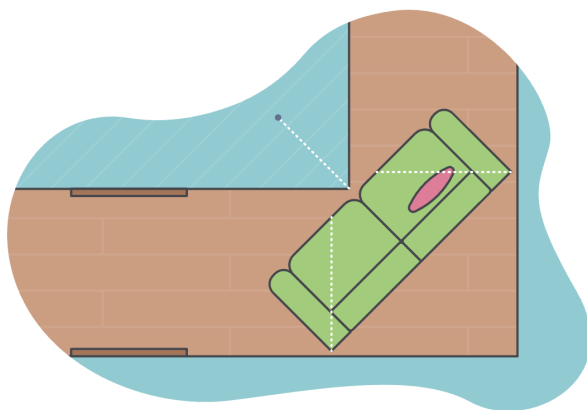
For me, this was the most difficult puzzle to date — partly because I wasn't sure that the problem had a solution and partly because I found it extremely difficult to imagine rotating and transposing a pill-shaped sofa around a tight corner. To overcome the limitations of my geometric imagination, I built a tiny model, which proved necessary for my solving the problem.

Puzzle #6: Moving Sofa Problem (Variation)

You have a hallway of width 1 that makes a 90° turn (an L-shaped corridor). Your sofa is *pill-shaped*. Specifically, it is a rectangle of length L and width 1, and attached to its two ends are semicircles of radius $\frac{1}{2}$.

1. What is the maximal length L of the rectangle part so that this pill-shaped sofa can be moved around the corner without colliding with the hallway walls?

$$L = \sqrt{2} - 1 \approx .414$$



Moving Sofa Problem for a Rectangular-Shaped Sofa

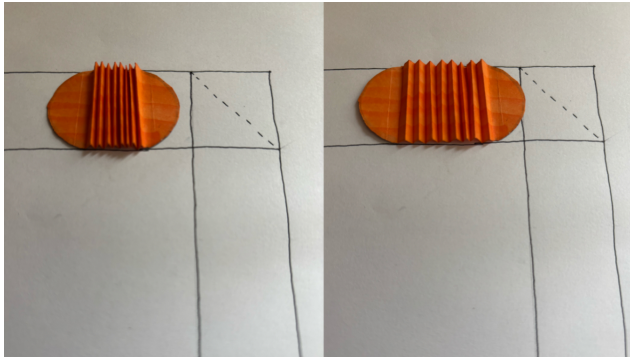
Shoutout (Romi) To my knowledge, my dear friend Romi is the only dog who subscribes to this newsletter. For centuries Cambridge philosophers have debated whether dogs can engage in syllogistic reasoning (See Logic for Dogs). In relation to this debate, I would like to note Romi's preternatural mathematical ability. Upon having the Hedgehog Theorem Puzzle read to her, she immediately identified her own bald spot, grasping that she herself is topologically equivalent to a ball. Rolling around and 'combing' herself, Romi provided a constructive proof by contradiction of the Hedgehog Theorem. Mathematics, as the vegans Kantians remind us, is within us all.



Romi in Her Office (2025)

Solution The maximal length $L = \sqrt{2} - 1 \approx .414$. This means that the largest pill-shaped sofa we can rotate along a tight corner has area $A = \frac{\pi}{4} + \sqrt{2} - 1 \approx 1.2$. Recall, the Gerver Sofa — the solution to the classical moving sofa problem — has area 2.22!

I discovered the key geometrical constraint by building a tiny model of a pill shaped sofa, which I designed like an accordion. This allowed me to test what effect slight changes in the length of the rectangle would have on the sofa's ability to round the corner.



My Tiny Pill-Shaped Sofa

After playing with my model for far too long, I grasped the key geometrical constraint on the sofa's length. In particular, it occurs when the sofa is rounding the corner and is rotated exactly 45° . This is the moment when both semi-circular ends are closest to the walls simultaneously. This position provides an upper bound for the sofa's length.

First, note that the corner of the corridor forms a unit square with diagonal length $\sqrt{2}$. This diagonal represents the maximum distance along which the rectangle can fit while rotated 45° .

The rectangle's width = 1, which is perpendicular to its length L . When rotated 45° , part of this width occupies space along the diagonal. Therefore, the maximum rectangle length along the diagonal is:

$$\begin{aligned} L_{max} &= \text{diagonal} - \text{width} \\ &= \sqrt{2} - 1 \\ &\approx .414 \end{aligned}$$

This ensures no part of the rectangle sticks outside the corridor. With this as the rectangle's length, we can easily check that the semi-circles of radius $\frac{1}{2}$ remain in the L-corridor as we make the turn. Indeed, they are exactly against the walls, as we would expect from an optimum solution. $\square$

Puzzle #7: Mutilated Chessboard Problem

The mutilated chessboard problem is a tiling puzzle first posed by Max Black in 1946. I gave this puzzle as a bonus question to the final exam in my summer Symbolic Logic course. About $\frac{2}{3}$ of the students got it correct. About the same number of students remarked it was a fun problem!

Consider a standard 8×8 chessboard with 64 squares colored alternately black and white. Suppose you have 32 dominoes, each of which is half black and half white. The board is initially covered completely by these 32 dominoes, where each domino covers exactly two adjacent squares such that the black half of the domino covers a black square and the white half covers a white square.

1. Now, suppose you remove the two opposite corner squares from the chessboard. Is it still possible to cover the remaining 62 squares exactly with 31 dominoes, each covering one black and one white square?
2. If yes, how? If not, why?



Standard Chess Board